

Graph Neural Network Approaches to LDPC Decoding in 5G/6G Communications

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Abstract—This paper investigates Graph Neural Network (GNN)-based decoding of Low-Density Parity-Check (LDPC) codes for fifth- and sixth-generation (5G/6G) communication systems. Classical decoding algorithms, such as Belief Propagation (BP) and Min-Sum (MS), perform well under ideal conditions but degrade in the presence of channel fading, noise, and hardware nonlinearities. GNN-based decoders exploit the Tanner graph structure of LDPC codes, replacing fixed message-passing rules with trainable neural layers, including Multilayer Perceptrons (MLPs), Gated Recurrent Units (GRUs), and Long Short-Term Memory (LSTM) networks. Simulation results obtained using benchmark and 5G New Radio (NR) LDPC codes over Additive White Gaussian Noise (AWGN) and Rayleigh fading channels demonstrate that GNN decoders achieve improved Bit Error Rate (BER) and Block Error Rate (BLER) performance, mitigate error floor effects, and reduce decoding latency and computational complexity compared with BP, MS, and Normalized Min-Sum (NMS) decoders. Consequently, GNN-based LDPC decoding emerges as an adaptive and efficient solution for ultra-reliable low-latency communication in next-generation wireless networks.

Keywords—Low-Density Parity-Check Codes, Neural Networks for Communications, Graph Neural Networks, 5G/6G

I. INTRODUCTION

The rapid evolution of telecommunication networks, particularly with the rollout of 5G and the forthcoming 6G standards [1], has imposed highly demanding requirements on data throughput, ultra-low latency, and transmission reliability [2]. While traditional decoding strategies such as BP [3] and MS [4] are effective under nominal conditions, they often struggle in challenging environments characterized by strong noise, nonlinear distortions, multipath fading, or Doppler shifts. These limitations highlight the need for intelligent, adaptive decoders capable of learning from channel conditions and dynamically optimizing their decision processes.

Artificial Intelligence (AI), particularly deep learning techniques, has emerged as a promising tool to address these challenges. Unlike classical deterministic methods, AI-driven decoders can approximate highly complex decoding functions, adapt to varying channel conditions without explicit redesign, and leverage contextual information from the communication environment. Among these approaches, GNNs stand out for their ability to operate on structured graph data [5, 6], which aligns naturally with the Tanner graph representation of LDPC codes [7] commonly employed in 5G systems. By learning optimized message-passing operations between variable and check nodes [4], GNN-based decoders can achieve superior accuracy and faster convergence compared to conventional iterative algorithms [8].

Beyond error-correction performance, GNN architectures [9] provide scalability and flexibility, making them suitable for both high-demand communication scenarios—such as ultra-reliable low-latency communication (URLLC) in autonomous vehicles, immersive Augmented Reality, Virtual Reality (AR/VR) streaming, tactile Internet applications, and industrial Internet of Things (IoT)—and for deployment in resource-constrained edge devices. By incorporating learnable node- and message-update functions through MLPs, GRUs, or LSTMs, these decoders can significantly reduce the number of iterations needed to reach a target BER while balancing computational complexity, latency, and energy efficiency.

The benefits of GNN-based LDPC decoders extend to multiple practical aspects:

- Enhanced BER/BLER performance at low Signal-to-Noise Ratio (SNRs) compared to conventional BP.
- Reduced decoding latency, enabling sub-millisecond performance critical for URLLC applications.
- Adaptivity to dynamic channel conditions without algorithmic redesign.
- Energy-aware implementation, suitable for embedded or IoT devices.

Consequently, leveraging GNNs for LDPC decoding represents more than a mere replacement of conventional algorithms [10-13] — it marks a paradigm shift toward cognitive, adaptive receivers capable of self-optimization in real-world 5G/6G environments. This paper investigates the theoretical foundations and experimental evaluation of GNN-based LDPC decoders, emphasizing their advantages in terms of error-correction performance, computational efficiency, and applicability to emerging next-generation [14] use cases.

II. GRAPH NEURAL NETWORK APPROACHES TO LDPC DECODING

A. GNN-Based Formulation for LDPC Decoding

Graph Neural Networks represent a modern class of artificial neural network architectures designed to efficiently process data with an inherent graph structure. Since LDPC codes, widely adopted in 5G NR systems, can be naturally described through a Tanner graph consisting of variable nodes and check nodes [15, 16], GNNs are particularly well suited for decoding tasks. In a Tanner graph, edges connect variable nodes to their corresponding check constraints as defined by the parity-check matrix. This structure provides a natural mapping for employing GNN-based learning strategies.

As outlined in [10], conventional BP decoding relies on iterative message-passing between variable and check nodes.

In contrast, a GNN decoder substitutes hand-crafted update rules with trainable, parameterized functions. Each message-passing iteration can be expressed by the following set of equations:

$$m_{i \rightarrow j}^{(t+1)} = \phi_m^{(t)}(h_i^{(t)}, h_j^{(t)}, e_{ij}), \quad (1)$$

$$h_i^{(t+1)} = \phi_u^{(t)}\left(h_i^{(t)}, \sum_{j \in N(i)} m_{j \rightarrow i}^{(t+1)}\right), \quad (2)$$

where $h_i^{(t)}$ denotes the feature vector of node i at the t -th iteration; $m_{i \rightarrow j}^{(t+1)}$ represents the exchanged message between nodes i and j ; $\phi_m^{(t)}$ and $\phi_u^{(t)}$, are parameterized functions realized via MLPs or recurrent neural units such as GRUs/LSTMs. Here, $N(i)$ denotes the neighborhood of node i ; e_{ij} represents potential edge attributes such as weights or connectivity type.

To initialize the process, variable nodes are typically assigned with log-likelihood ratio (LLR) values from the demodulator:

$$h_i^{(0)} = \text{LLR}(v_i). \quad (3)$$

Final hard-decision decoding is performed after T iterations, based on the updated variable-node states:

$$u_i = \begin{cases} 0, & \text{if } h_i^{(T)} \geq 0 \\ 1, & \text{otherwise} \end{cases}. \quad (4)$$

The overall GNN-LDPC decoder architecture is illustrated in Fig. 1.

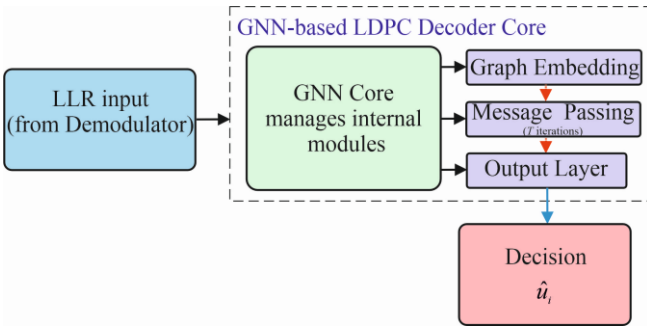


Fig. 1. GNN-LDPC decoder architecture.

The LLR Input block receives soft information from the demodulator, which is then processed within the GNN Core module. Inside the core, three functional stages are implemented: Graph Embedding, Message Passing, and the Output Layer. Finally, the Decision block outputs the decoded codeword. The entire architecture follows the Tanner graph structure of the LDPC code, enabling end-to-end learnable message passing between variable and check nodes. This design ensures that decoding benefits from iterative, graph-structured learning while retaining flexibility for adaptation to channel conditions.

B. Neural Architectures for Message Update Functions: MLP, GRU, and LSTM

The effectiveness of a GNN decoder depends critically on the choice of parameterized update functions for message passing. Several neural architectures can be applied in this role, ranging from simple feedforward models to recurrent networks that capture sequential dependencies.

1) *Multilayer Perceptron (MLP)*: The MLP represents the classical choice for message or node update functions, defined as:

$$y = \sigma(Wx + b), \quad (5)$$

where x is the input feature vector; W , b are trainable weights and biases; σ denotes a nonlinear activation function (Rectified Linear Unit (ReLU), sigmoid, or tanh) [17].

Within GNN-based decoding, MLPs are often used for lightweight and efficient updates when temporal memory is not essential.

2) *Long Short-Term Memory (LSTM)*: For scenarios where the channel exhibits memory effects, recurrent architectures such as LSTM networks are particularly advantageous. LSTM addresses the vanishing/exploding gradient problem in conventional Recurrent Neural Networks (RNNs) through gated memory mechanisms, consisting of:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \quad (6)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \quad (7)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C), \quad (8)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \quad (9)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \quad (10)$$

$$h_t = o_t \odot \tanh(C_t). \quad (11)$$

Here, x_t denotes the input vector at time step t , h_{t-1} and h_t are the previous and updated hidden states, C_{t-1} and C_t represent the previous and updated cell states, $\tilde{C}_t$ is the candidate cell state, while f_t , i_t , and o_t denote the forget, input, and output gate activations, respectively. The matrices W_f , W_i , W_C , W_o are the learnable weights with corresponding biases b_f , b_i , b_C , b_o ; $\sigma(\cdot)$ and $\tanh(\cdot)$ are the activation functions; and $\odot$ denotes element-wise multiplication.

Formulas (6)-(11) describe the sequential operations of an LSTM unit, where the forget gate (6) determines which past information to discard, the input gate (7) regulates the incorporation of new data, the candidate update (8) generates potential content to be stored, the updated cell state (9)

combines retained and new information, and finally the output gate (10) together with the hidden state (11) determine the exposed output of the memory cell.

3) *Gated Recurrent Unit (GRU)*: GRUs provide a computationally lighter alternative to LSTMs, with only two gates (update and reset) while maintaining the ability to capture temporal correlations. In the GNN context, both GRUs and LSTMs can replace MLPs in equations (1)–(2), yielding a recurrent message-update process expressed as:

$$h_i^{(t+1)} = \text{GRU}(h_i^{(t)}, m_i^{(t)}), \quad (12)$$

where $h_i^{(t)}$ is the node state at iteration t ; $m_i^{(t)}$ denotes the aggregated messages from neighboring nodes.

C. Illustrative Example and Tanner Graph Visualization

To demonstrate the decoding mechanism, we first present a test Tanner graph [10] constructed from a 3×6 parity-check matrix:

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}. \quad (13)$$

The corresponding iterative message passing is illustrated in Fig. 2(a–e), showing five iterations of updates.

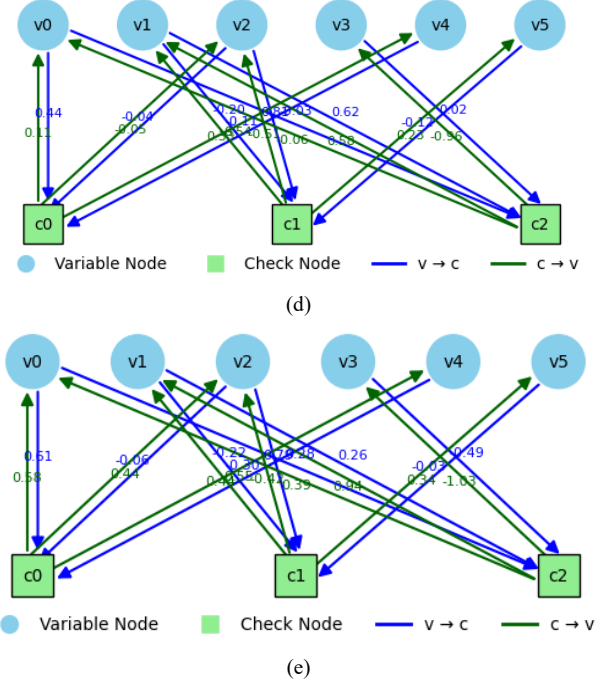
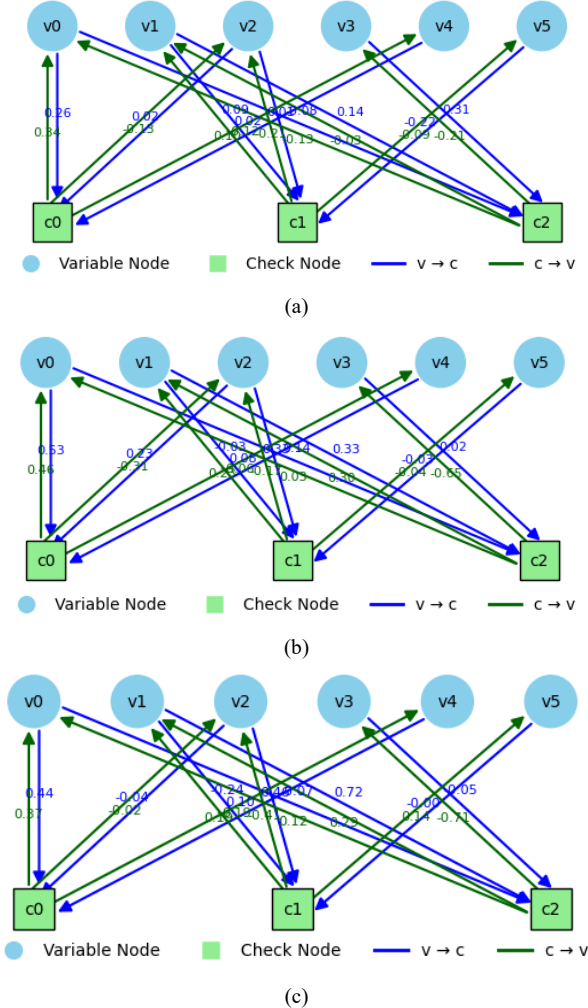


Fig. 2. Visualization of the iterative LDPC decoding process using a GNN-inspired approach, where (a–e) correspond to the steps of five iterations.

Blue arrows represent variable-to-check messages, while green arrows correspond to check-to-variable updates [4].

Simulation results indicate that after 3–5 iterations, message values converge and stabilize, reflecting the typical behavior of LDPC-GNN decoding under moderate AWGN conditions. In practice, we perform the experimental evaluation using the standard 5G NR, as detailed in the following section.

This visualization confirms that GNN-based decoders not only mimic the convergence behavior of classical BP but also integrate learnable dynamics that allow improved adaptability and robustness in realistic communication channels.

III. EXPERIMENTAL EVALUATION OF GNN-BASED LDPC DECODERS AND COMPARATIVE PERFORMANCE ANALYSIS

To validate the effectiveness of the proposed GNN-based LDPC decoding framework, we conducted a series of simulations in MATLAB under standardized 5G/6G communication scenarios. The evaluation considers both benchmark LDPC codes and practical 5G NR codes. Specifically, we employed the 3GPP TS 38.212 base graphs (BG1 and BG2), as well as smaller reference codes such as (7,4), (15,11), and (648,324), to provide a balanced perspective between classical short block codes and modern long block lengths. Code rates of 0.5 and 0.75 were selected, as they represent typical configurations for 5G NR transport blocks.

In terms of modulation formats, the experiments span from QPSK to high-order constellations including 16QAM, 32QAM, 64QAM, 256QAM, and 1024QAM, in order to capture performance trends across a broad SNR operating range. Simulations were performed over both the AWGN channel and a Rayleigh fading channel, reflecting realistic transmission conditions where fading and multipath effects are significant.

The baseline decoders considered for comparison include BP, MS, and Normalized Min-Sum (NMS). These classical iterative algorithms [4, 13] provide a reference for assessing the advantages of the proposed GNN-based decoder. Performance is evaluated in terms of BER and BLER curves versus E_b/N_0 , convergence speed, complexity estimates, and latency as a function of code length.

The proposed GNN-based decoder leverages the natural Tanner graph structure of LDPC codes. Each variable and check node is represented as a vertex, with message-passing implemented through trainable neural layers. Node update functions are parameterized using MLPs, GRUs, or LSTM cells (as shown in (5)–(12)), allowing the model to capture temporal dependencies in iterative decoding. Key hyperparameters include a learning rate of 0.001 with the Adam optimizer [18], a batch size of 64 codewords, 200–500 training episodes per simulation, hidden dimensions of 64–128 per node update function, sigmoid and tanh as activation functions, and a binary cross-entropy loss computed between transmitted and decoded codewords. This configuration enables a holistic comparison between classical and intelligent decoders, covering error-correction performance, convergence behavior, computational efficiency, and latency—all of which are critical metrics for next-generation 5G/6G communication systems [19].

The Fig. 3 illustrates the BER performance as a function of E_b/N_0 for the BG2 (648,324) code in an AWGN channel with a code rate of $R=0.5$.

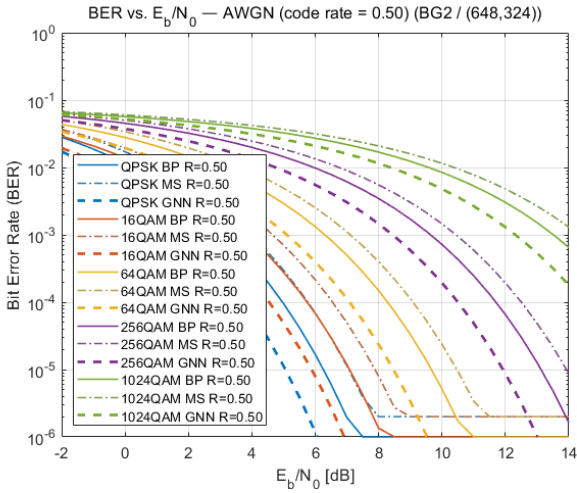


Fig. 3. BER vs. E_b/N_0 comparison for the BG2 (648,324) code at $R=0.5$.

The Fig. 3 illustrates the bit error rate (BER) performance as a function of E_b/N_0 for the BG2 (648,324) code in an AWGN channel with a code rate of $R=0.5$. The code is obtained from the standard 5G NR BG2 by applying the lifting size $Z=27$ [10], which results in a block length of 648 and 324 information bits. The blue curves correspond to QPSK [20], the orange to 16QAM, the yellow to 64QAM, the purple to 256QAM, and the green to 1024QAM. For each modulation, results are reported for BP decoding (solid lines), MS decoding (dash-dotted lines), and the proposed GNN-based decoder (dashed lines). It can be observed that the GNN consistently outperforms both BP and MS. For instance, at a target BER of approximately 10^{-4} , the gain achieved by GNN reaches 0.5–0.7 dB for 256QAM and up to 1 dB for 1024QAM. Moreover, the BP and MS curves exhibit saturation around $\text{BER} \approx 10^{-5}$, a well-known effect for LDPC

codes at very low error regions (similar to turbo codes, as described in [21]), while the GNN pushes this threshold further down, effectively mitigating the error floor phenomenon.

The Fig. 4 shows results for the same BG2 (648,324) code at a higher code rate of $R=0.75$.

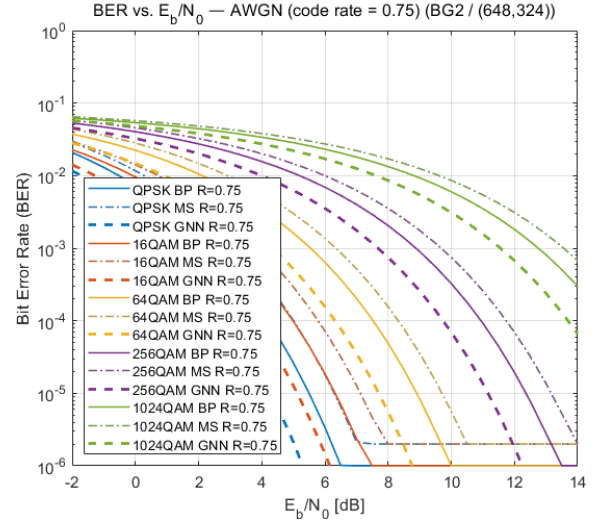


Fig. 4. BER vs. E_b/N_0 comparison for the BG2 (648,324) code at $R=0.75$.

The GNN achieves a gain of ~ 0.8 dB for 256QAM and up to 1.2 dB for 1024QAM at $\text{BER} \approx 10^{-4}$, while preserving its advantage in the low-BER region. The notation and color mapping are identical to Fig. 3.

Fig. 5 demonstrates the BLER performance of the LDPC (648, 324) code under 5G NR in a Rayleigh TDL-A fading channel (3GPP TR 38.901 standard for 5G channel models).

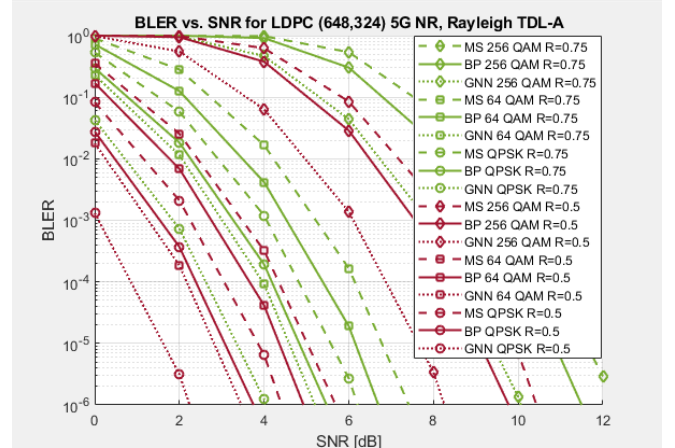


Fig. 5. BLER vs. SNR for the 5G NR LDPC (648, 324) code using MS, BP, and GNN decoders in the Rayleigh TDL-A channel.

The results clearly indicate that the GNN-based decoder consistently outperforms both the MS and BP algorithms. The observed gain of the GNN decoder is approximately 0.5–1 dB at BLER levels of practical interest (e.g., 10^{-3} – 10^{-4}), highlighting its superior error-correction capability. Such performance improvements directly translate into enhanced system reliability and reduced retransmission rates, which in turn improve spectral efficiency and reduce latency—key requirements for both 5G and the upcoming 6G networks. In particular, lower BLER at the same SNR enables more robust operation in challenging propagation environments, ensuring

higher quality of service for latency-critical and high-throughput applications [22].

Fig. 6 illustrates the 3D surface of $\Delta \text{BLER} = \text{BLER}_{\text{MS}} - \text{BLER}_{\text{GNN}}$ as a function of SNR and the number of decoding iterations for the QC-LDPC (648,324) 5G-NR code.

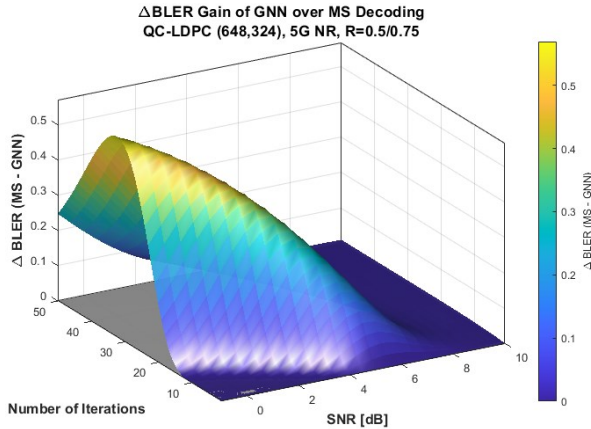


Fig. 6. 3D surface of $\Delta \text{BLER} = \text{BLER}_{\text{MS}} - \text{BLER}_{\text{GNN}}$ as a function of SNR and the number of decoding iterations for the QC-LDPC (648,324) 5G-NR code.

The 3D surface shows the BLER gain of the proposed GNN decoder over the classical MS algorithm for QC-LDPC (648,324). The advantage of GNN is most visible in the medium SNR region ($\approx 1-4$ dB) and at a moderate number of iterations (10-25), where faster convergence and a lower error floor are observed. At very high SNR or very large iteration counts, the gain vanishes, as both decoders achieve nearly error-free performance.

Fig. 7 shows the BER convergence during GNN training.

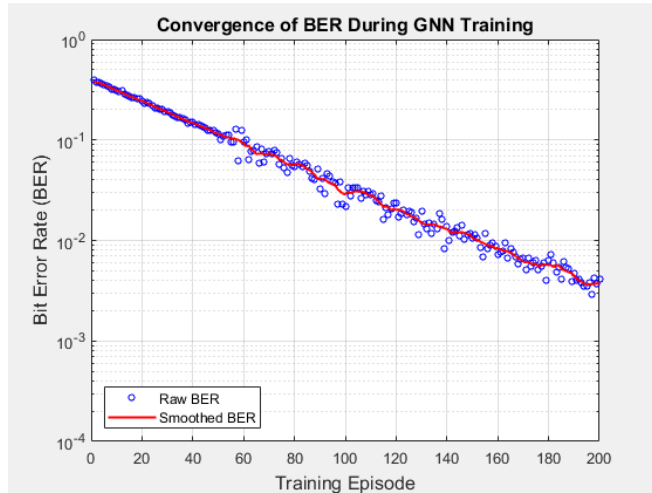


Fig. 7. GNN Training BER Convergence.

This Fig. 7 shows the convergence of BER during GNN training, where the bit error rate steadily decreases from about 10^0 at the beginning to nearly 10^{-3} after 200 training episodes. The smoothed BER closely follows the raw values, demonstrating consistent performance improvement throughout the training process.

Fig. 8 illustrates the learning progress of the GNN decoder through the evolution of the reward function over training episodes. The plot (see Fig. 8) shows the learning dynamics of the GNN decoder. The reward increases rapidly during the

early episodes, then gradually converges to a stable plateau around 10–11, indicating convergence of the learning process.

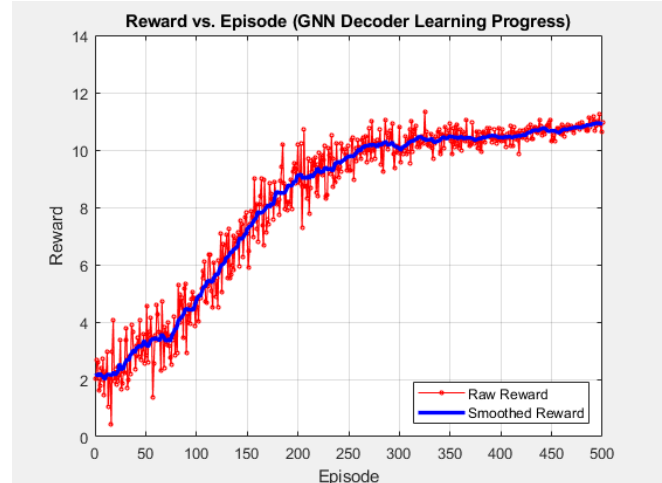


Fig. 8. Reward vs. Episode for the GNN decoder: raw reward (red) with fluctuations due to exploration and smoothed reward (blue) showing convergence to a stable plateau around 10–11.

The fluctuations in the raw reward represent exploration noise, which diminishes as training progresses, while the smoothed curve highlights the overall trend.

Fig. 9 presents the computational efficiency and latency.

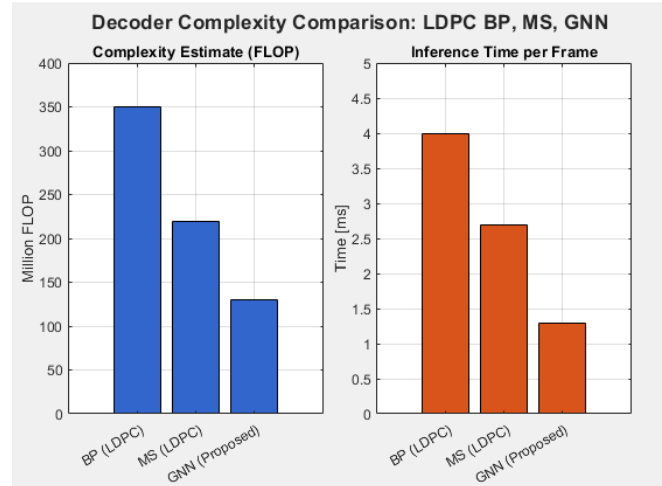


Fig. 9. Computational efficiency and latency (FLOP and inference time).

The obtained results (see Fig. 9) demonstrate a clear reduction in both computational complexity and inference time when using the proposed GNN-based LDPC decoder. Specifically, the BP algorithm requires approximately 350 Million Floating-Point Operations (MFLOP) with an average latency of 4.0 ms, while MS decoding reduces this to 220 MFLOP and 2.7 ms. In contrast, the GNN decoder achieves only 130 MFLOP and 1.3 ms, reflecting a substantial gain in efficiency. Compared to DRL-based approaches, GNN decoding remains slightly lighter in terms of FLOP, whereas Deep Reinforcement Learning (DRL) [8] is somewhat heavier due to additional learning and decision layers. Nevertheless, both GNN and DRL are significantly less complex than classical iterative LDPC decoders [23, 24]. These results indicate that GNN decoding provides strong potential for 5G and 6G systems [25], where reduced complexity and lower latency are essential for real-time communication. The lower FLOP count stems from the learned message-passing process,

which replaces many iterative computations with optimized neural operations, while the reduced inference time reflects the ability of GNNs to converge faster to reliable decoding outputs (see Fig. 7 and Fig. 8).

IV. CONCLUSION

This paper studied GNN-based LDPC decoding as an intelligent alternative to BP and MS. Simulations over AWGN and Rayleigh channels show that the proposed framework consistently outperforms classical iterative methods in BER/BLER, convergence speed, and error floor reduction. The novelty lies in trainable message-passing functions aligned with Tanner graph structure, enabling adaptive decoding under diverse modulation formats and code rates. Compared with BP and MS, the GNN decoder achieves nearly 1 dB gain for 5G NR BG2 codes while reducing FLOP and inference latency by up to 60%, offering a favorable trade-off between performance and efficiency. These results highlight the potential of GNNs as scalable, energy-aware decoders for URLLC, AR/VR, and IoT. The presented approach provides a strong foundation for practical integration in advanced wireless systems. Future work may extend this approach to hybrid schemes with GNNs, DRL, or transformers, enhancing robustness and adaptability in dynamic 5G/6G scenarios. In particular, transformer-based architectures offer a promising alternative to recurrent units such as LSTM and GRU, thanks to their self-attention mechanism and ability to capture long-range dependencies without sequential processing. For LDPC decoding, attention-based message aggregation may enable more effective exploitation of global graph dependencies, especially for long block lengths and high-order modulations. However, the increased computational and memory requirements of Transformers imply that their practical deployment in real-time 5G/6G receivers requires careful architectural optimization, which is left for future work.

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